

Somaiya Vidyavihar University
K J Somaiya School of Engineering
ISE Syllabus and Scheme for Internal Assessment (IA)

Department	Electronics and Telecommunication Engineering		
Program	EXTC	Year	SY B.Tech
Academic year:	January 2026 - April 2026	Course Code and Name	Communication Systems (216U03C403)
Semester:	IV		
Faculty In-charge	Mr. Kartik Patel		

Distribution of 20 Marks	The student will be evaluated based on weekly tests. If any of the tests is not completed / submitted / shown / evaluated, then the marks assigned for that test will be ZERO.			
	Internal Assessment-I (IA-I) and Internal Assessment-II (IA-II): Weekly Tests			
	There will be weekly test on every MONDAY (as per the following table) during the lecture. The test will have the questions/activity covering portion covered in the previous week. The questions will be logical/MCQs/short answer/long answer etc. The time limit will be announced on the same day depending of the nature/type of the questions/activity. Best 8 tests marks out of 11 will be considered for calculating IA marks out of 20.			
	12/01/2026		02/03/2026	
	19/01/2026		23/03/2026	
	02/02/2026		30/03/2026	
	09/02/2026		06/04/2026	
	23/02/2026		13/04/2026	
	20/04/2026			
	There will be NO repetition for any of the weekly tests in case students miss/remain absent because of any reason.			
Additional Internal Assessment: Reflection Journal of Lecture				
Each student is required to maintain a Reflection Journal for every lecture, clearly recording the date, day, and topics covered, along with the student’s self-reflection on the lecture. Students are advised to maintain a separate notebook exclusively for this purpose. The Reflection Journal will be verified at the end of the semester, after the IA marks have been finalized. For students who obtain lower IA scores (as determined by the course instructor), a properly maintained and complete Reflection Journal may be considered for the award of limited additional marks, subject to the discretion of the course in charge. Please note that the Reflection Journal will not be considered for awarding full marks, highest marks, grade improvement, or significant enhancement of IA scores. The criteria and extent of mark consideration, if any, will be decided by the course in charge at the end of the semester. The journal must be neatly maintained, complete, and contain genuine self-reflections for each lecture.				
Sr. No.	Task	Module No.	Schedule	Mode of Conduction
1.	Weekly tests	Module No. 1 to 5	As given in table above	Offline in classroom (paper and pen)
Final IA marks will be out of 20: Best of 8 tests marks out of 11				

Date: 05/01/2026

Faculty in-charge: Mr. Kartik Patel

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Syllabus for In Semester Examination (ISE) - Maximum marks 30

Module No.	Topics		CO Mapping
1	Fundamentals of Electronic Communication Systems		CO 2 CO 4
	1.1	Electromagnetic spectrum, Basic block diagram of communication system, Concept of modulation and demodulation	
	1.2	Sources of noise, signal to noise ratio, noise factor, noise figure, noise temperature and related numerical.	
2	Amplitude Modulation and Demodulation		CO 1 CO 2 CO 4
	2.1	AM Signal: Waveform, mathematical representation, spectrum, modulation index, bandwidth, voltage distribution, power calculations and related numerical.	
	2.2	DSB-FC system: Modulation, low level and high level modulation; Demodulation: envelope detector, practical diode detector	
	2.3	DSB-SC system: Modulation, Balanced modulator, Ring modulator; Demodulation: Coherent Detector, Product detector	
3	Basic Features of Angle Modulation and Demodulation		CO 1 CO 2 CO 4
	3.1	Angle Modulation: Waveform, mathematical representation, modulation index, frequency and phase deviation, spectrum, bandwidth; Narrow Band FM, Wide Band FM	
	3.2	FM system: Modulation, Direct methods – Varactor diode modulator, FET reactance modulator; Indirect method – Armstrong's method; Noise triangle in FM; Pre-emphasis and de-emphasis; FM transmitter	

Calculation of final CA marks (out of 50):

Internal Assessments (IA)	In Semester Examination (ISE)
Weekly Tests	
Maximum Marks: 20	Maximum Marks: 30
Best 8 tests marks out of 11	Final ISE Marks (out of 30 Marks)
Final CA Marks (out of 50) = IA Marks (out of 20) + ISE Marks (out of 30)	

Date: 03/01/2026

Faculty in-charge: Mr. Kartik Patel